Natural Language Processing HW 2

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1. **Manipulating Probabilities**

   a. Since $Y \subseteq Z$, we can say that

   $$\begin{aligned} & p(Z) = p(Y) + p(Z - Y) \\ \implies & p(Z - Y) = p(Z) - p(Y) \end{aligned}$$

   We know that probabilities cannot be negative, thus

   $$\begin{aligned} & p(Z - Y) \geq 0 \\ \implies & p(Z) - p(Y) \geq 0 \\ \implies & p(Z) \geq p(Y) \end{aligned}$$

   Thus proved.

   b. We know that $p(X|Z)$ can be rewritten as $\frac{p(X \cap Z)}{p(Z)}$.

   We also know that

   $$\begin{aligned} & X \cap Z \subseteq Z \\ \implies & p(X \cap Z) \leq p(Z) \quad \text{(from (a))} \\ \implies & \frac{p(X \cap Z)}{p(Z)} \leq 1 \\ \implies & p(X|Z) \leq 1 \end{aligned}$$

   Also, from the fact that probabilities cannot be negative, we know that

   $$p(X|Z) \geq 0$$

   This gives us $0 \leq p(X|Z) \leq 1$, which means that conditional probabilities fall within the rnage $[0, 1]$ as well.

   Thus proved.

   c. We can say that

   $$\mathcal{E} = \mathcal{E} \cup \emptyset,$$

where $\mathcal{E}$ and $\emptyset$ are disjoint sets - i.e., $\mathcal{E} \cap \emptyset = \emptyset$.

Given that these two sets are disjoint, we also know from the axioms that

$$\begin{aligned} & p(\mathcal{E} \cup \emptyset) = p(\mathcal{E}) + p(\emptyset) \\ \implies & p(\mathcal{E}) = 1 + p(\emptyset) \\ \implies & 1 = 1 + p(\emptyset) \\ \implies & p(\emptyset) = 0 \end{aligned}$$

Thus proved.

d. Since $\bar{X}$ denotes $\mathcal{E} - X$, we can say that

$$\mathcal{E} = X \cup \bar{X}$$

We also know that $\bar{X}$ and X are disjoint sets, given that one is the complement of another. Thus,

$$\begin{aligned} & p(\mathcal{E}) = 1 \\ \implies & p(X \cup \bar{X}) = 1 \\ \implies & p(X) + p(\bar{X}) = 1 \\ \implies & p(X) = 1 - p(\bar{X}) \end{aligned}$$

Thus proved.

e. Let *singing* be denoted by S, and *rainy* by R. We can thus rewrite p(singing AND rainy | rainy) as $p(S \cap R|R)$, which we further simplify as follows:

$$\begin{aligned} & p(S \cap R|R) \\ =& \frac{p((S \cap R) \cap R)}{p(R)} \\ =& \frac{p(S \cap (R \cap R))}{p(R)} \quad \text{(since intersection is associative)} \\ =& \frac{p(S \cap R)}{p(R)} \\ =& p(S|R) = p(singing|rainy) \end{aligned}$$

Thus proved.

f. We know that $X \cup \bar{X} = \mathcal{E}$, which implies

$$\mathcal{E} \cap Y = (X \cup \bar{X}) \cap Y = Y.$$

Taking probabilities, we get

$$\begin{aligned}
& p((X \cup \bar{X}) \cap Y) = p(Y) \\
\implies & p((X \cap Y) \cup (\bar{X} \cap Y)) = p(Y) \\
\implies & p(X \cap Y) + p(\bar{X} \cap Y) = p(Y) \quad \text{(since } X \cap Y \text{ and } \bar{X} \cap Y \text{ are disjoint)} \\
\implies & \frac{p(X \cap Y) + p(\bar{X} \cap Y)}{p(Y)} = 1 \\
\implies & \frac{p(X \cap Y)}{p(Y)} = 1 - \frac{p(\bar{X} \cap Y)}{p(Y)} \\
\implies & p(X|Y) = 1 - p(\bar{X}|Y)
\end{aligned}$$

Thus proved.

g.

$$\begin{aligned}
& \big(p(X \mid Y) \cdot p(Y) + p(X \mid \bar{Y}) \cdot p(\bar{Y})\big) \cdot \frac{p(\bar{Z} \mid X)}{p(\bar{Z})} \\
= & \left(\frac{p(X \cap Y)}{p(Y)} \cdot p(Y) + \frac{p(X \cap \bar{Y})}{p(\bar{Y})} \cdot p(\bar{Y})\right) \cdot \frac{p(\bar{Z} \cap X)}{p(X) \cdot p(\bar{Z})} \\
= & \big(p(X \cap Y) + p(X \cap \bar{Y})\big) \cdot \frac{p(\bar{Z} \cap X)}{p(X) \cdot p(\bar{Z})} \\
= & p(X) \cdot \frac{p(\bar{Z} \cap X)}{p(X) \cdot p(\bar{Z})} \\
= & \frac{p(\bar{Z} \cap X)}{p(\bar{Z})} \\
= & p(X|\bar{Z})
\end{aligned}$$

h. We know that $p(singing\ OR\ rainy)$ can be rewritten as

$$p(singing\ \cup\ rainy) = p(singing) + p(rainy) - p(singing \cap rainy)$$

Thus, $p(singing\, OR\, rainy) = p(singing) + p(raining)$ is only true when $p(singing \cap rainy) = 0$.

i. We know that

$$p(singing \mid rainy) = \frac{p(singing\ \cap\ rainy)}{p(rainy)}.$$

We also know that

$$p(singing \mid rainy) = p(singing)$$

only when the two events are independent.

Thus, if the two events are independent, we get

$$\begin{aligned}
p(singing) &= \frac{p(singing\ \cap\ rainy)}{p(rainy)} \\
\implies p(singing) \cdot p(rainy) &= p(singing\ \cap\ rainy)
\end{aligned}$$

Thus, $p(singing) \cdot p(rainy) = p(singing \ \cap \ rainy)$ when the *singing* and *rainy* events are independent.

j. We know that

$$\begin{aligned} & p(X \mid Y) = 0 \\ \implies & \frac{p(X \cap Y)}{p(Y)} = 0 \\ \implies & p(X \cap Y) = 0. \end{aligned}$$

Now, analyzing $p(X \mid Y, Z)$:

$$\begin{aligned} & p(X \mid Y, Z) \\ = & \frac{p(X \cap (Y \cap Z))}{p(Y \cap Z)} \\ = & \frac{p((X \cap Y) \cap Z)}{p(Y \cap Z)}. \end{aligned}$$

Since

$$(X \cap Y) \cap Z \subseteq X \cap Y,$$

by part (a),

$$p((X \cap Y) \cap Z) \leq p(X \cap Y) = 0.$$

Since probabilities are nonnegative,

$$p((X \cap Y) \cap Z) = 0.$$

Therefore,

$$\begin{aligned} p(X \mid Y, Z) &= \frac{0}{p(Y \cap Z)} \\ &= 0. \end{aligned}$$

Thus proved.

k. We know that

$$p(W \mid Y) = 1.$$

By the definition of conditional probability,

$$\frac{p(W \cap Y)}{p(Y)} = 1,$$

so

$$p(W \cap Y) = p(Y).$$

This means that all of the probability mass in Y lies inside W. Therefore, the part of Y outside W must have probability 0:

$$p(Y \cap \overline{W}) = 0.$$

Now we want to see what happens when we additionally condition on Z. The part of $Y \cap Z$ that lies outside W is

$$Y \cap Z \cap \overline{W}.$$

Since

$$Y \cap Z \cap \overline{W} \subseteq Y \cap \overline{W},$$

by part (a),

$$p(Y \cap Z \cap \overline{W}) = 0.$$

Now split $Y \cap Z$ into the part inside W and the part outside W:

$$p(Y \cap Z) = p(W \cap Y \cap Z) + p(\overline{W} \cap Y \cap Z).$$

Since

$$p(\overline{W} \cap Y \cap Z) = 0,$$

we get

$$p(Y \cap Z) = p(W \cap Y \cap Z).$$

Therefore,

$$\begin{aligned} p(W \mid Y, Z) &= \frac{p(W \cap Y \cap Z)}{p(Y \cap Z)} \\ &= \frac{p(Y \cap Z)}{p(Y \cap Z)} \\ &= 1. \end{aligned}$$

Thus proved.

2. **Red car, Blue car**

   a. We can relate these terms using Bayes' Theorem:

   $$p(Actual = blue \mid Claimed = blue) = \frac{p(Claimed = blue \mid Actual = blue) \cdot p(Actual = blue)}{p(Claimed = blue)}$$

   b. The event $Actual = blue$ is the hypothesis we are evaluating, and the event $Claimed = blue$ is the evidence we have observed.

   $p(Actual = blue)$ is the prior probability of the hypothesis, $p(Claimed = blue \mid Actual = blue)$ is the likelihood, and $p(Actual = blue \mid Claimed = blue)$ is the posterior probability of the hypothesis.

   c. From the question,

   $$p(Actual = blue) = 0.1.$$

   From the question again,

   $$p(Claimed = blue \mid Actual = blue) = 0.8,$$

since witnesses are believed to be 80% reliable in reporting car color, regardless of the actual car color.

We will calculate $p(Actual = blue \mid Claimed = blue)$ using Bayes Theorem, which gives us

$$p(Actual = blue \mid Claimed = blue) = \frac{p(Claimed = blue \mid Actual = blue) \cdot p(Actual = blue)}{p(Claimed = blue)}$$

The only value we do not have here is $p(Claimed = blue)$, which we can obtain by enumerating all possible cases.

Since 10% of cars that hit pedestrians are blue, 8% of them will be claimed to be blue, while 2% of them will be claimed to be red. Similarly, since 90% of cars that hit pedestrians are red, 72% of them will be claimed to be red, while 18% of them will be claimed to be blue.

This means that

$$p(Claimed = blue) = 0.08 + 0.18 = 0.26.$$

Thus, substituting this value into Bayes Theorem:

$$\begin{aligned} & p(Actual = blue \mid Claimed = blue) \\ =& \frac{p(Claimed = blue \mid Actual = blue) \cdot p(Actual = blue)}{p(Claimed = blue)} \\ =& \frac{0.8 \cdot 0.1}{0.26} \\ =& \frac{0.08}{0.26} \approx 0.308 \end{aligned}$$

The judge should care about the posterior probability of the hypothesis, or $p(Actual = blue \mid Claimed = blue)$, since the witness is already testifying that a blue car hit the pedestrian, and what matters is how likely it was that the car was actually blue.

d. We have to prove that

$$p(A|B, Y) = \frac{p(B|A, Y) \cdot p(A|Y)}{p(B|Y)}.$$

Taking the RHS and simplifying,

$$\begin{aligned}
&\frac{p(B|A,Y)\cdot p(A|Y)}{p(B|Y)}\\
=&\frac{\frac{p(B\cap A\cap Y)}{p(A\cap Y)}\cdot\frac{p(A\cap Y)}{p(Y)}}{\frac{p(B\cap Y)}{p(Y)}}\\
=&\frac{\frac{p(B\cap A\cap Y)}{p(Y)}}{\frac{p(B\cap Y)}{p(Y)}}\\
=&\frac{p(B\cap A\cap Y)}{p(B\cap Y)}\\
=&\frac{p(A\cap(B\cap Y))}{p(B\cap Y)}\\
=&p(A|B\cap Y)\\
=&p(A|B,Y)
\end{aligned}$$

Thus proved.

e. We have to prove that

$$p(A|B,Y)=\frac{p(B|A,Y)\cdot p(A|Y)}{p(B|A,Y)\cdot p(A|Y)+p(B|\bar{A},Y)\cdot p(\bar{A}|Y)}.$$

First we simplify the denominator of the RHS,

$$\begin{aligned}
&p(B|A,Y)\cdot p(A|Y)+p(B|\bar{A},Y)\cdot p(\bar{A}|Y)\\
=&\frac{p(B\cap A\cap Y)}{p(A\cap Y)}\cdot\frac{p(A\cap Y)}{p(Y)}+\frac{p(B\cap\bar{A}\cap Y)}{p(\bar{A}\cap Y)}\cdot\frac{p(\bar{A}\cap Y)}{p(Y)}\\
=&\frac{p(B\cap A\cap Y)}{p(Y)}+\frac{p(B\cap\bar{A}\cap Y)}{p(Y)}\\
=&\frac{p(B\cap Y)}{p(Y)}\quad(\text{since }(B\cap A)+(B\cap\bar{A})=B)\\
=&p(B|Y)
\end{aligned}$$

Substituting into the RHS, we get

$$\frac{p(B|A,Y)\cdot p(A|Y)}{p(B|Y)},$$

which we already know is equal to the LHS, $p(A|B,Y)$, from (d).

Thus proved.

f.

$$\begin{aligned}
&p(Actual=blue\mid Claimed=blue,Baltimore)\\
=&\frac{p(Claimed=blue\mid Actual=blue,Baltimore)\cdot p(Actual=blue\mid Baltimore)}{p(Claimed=blue\mid Actual=blue,Baltimore)\cdot p(Actual=blue\mid Baltimore)+p(Claimed=blue\mid NOT\ Actual=blue,Baltimore)\cdot p(NOT\ Actual=blue\mid Baltimore)}\\
=&\frac{0.8\cdot 0.1}{0.8\cdot 0.1+0.2\cdot 0.9}
\end{aligned}$$

3. **Beaver Cries**

a. $\sum_{cry} p(cry \mid situation) = 1$

b.

p(cry, situation)	Predator!	Timber!	I need help!	TOTAL
bwa	0	0	0.64	0.64
bwee	0	0	0.08	0.08
kiki	0.2	0	0.08	0.28
TOTAL	0.2	0	0.8	1

c. i. $p(predator \mid kiki)$

ii. $\frac{p(predator \cap kiki)}{p(kiki)}$

iii.

$$\begin{aligned} &\frac{p(predator \cap kiki)}{p(kiki)} \\ =&\frac{0.2}{0.28} \\ \approx& 0.714 \end{aligned}$$

iv. $\frac{p(kiki|predator) \cdot p(predator)}{p(kiki|predator) \cdot p(predator) + p(kiki|timber) \cdot p(timber) + p(kiki|help) \cdot p(help)}$

v.

$$\begin{aligned} &\frac{p(kiki \mid predator) \cdot p(predator)}{p(kiki \mid predator) \cdot p(predator) + p(kiki \mid timber) \cdot p(timber) + p(kiki \mid help) \cdot p(help)} \\ =&\frac{1 \cdot 0.2}{1 \cdot 0.2 + 0.3 \cdot 0 + 0.1 \cdot 0.8} \\ =&\frac{0.2}{0.28} \\ \approx& 0.714 \end{aligned}$$

4. **Language Models**

a.

$$\begin{aligned} &p(w_1 w_2 w_3 w_4) \\ =&p(w_1 \mid BOS, BOS) \cdot p(w_2 \mid BOS, w_1) \cdot p(w_3 \mid w_1, w_2) \cdot p(w_4 \mid w_2, w_3) \cdot p(EOS \mid w_3, w_4) \\ =&\frac{c(BOSBOSw_1)}{c(BOSBOS)} \cdot \frac{c(BOSw_1w_2)}{c(BOSw_1)} \cdot \frac{c(w_1w_2w_3)}{c(w_1w_2)} \cdot \frac{c(w_2w_3w_4)}{c(w_2w_3)} \cdot \frac{c(w_3w_4EOS)}{c(w_3w_4)} \end{aligned}$$

$c(BOS, BOS)$ counts the number of sentences that start with BOS, BOS,
$c(BOS, BOS, i)$ counts the number of sentences that start with BOS, BOS, i, and
$c(bagpipe, music, EOS)$ counts the number of sentences that end with $bagpipe, music, EOS$.

b. $p(\vec{w})$ is low for $\vec{w} = (do, you, think, the)$ because "do you think the" is not a complete sentence. For a trigram language model,

$$\begin{aligned} p(\vec{w}) = p(do \mid BOS, BOS) \cdot p(you \mid BOS, do) \cdot p(think \mid do, you) \\ \cdot p(the \mid you, think) \cdot p(EOS \mid think, the), \end{aligned}$$

in which $p(EOS \mid think, the)$ is really, really low, which causes the overall probability of $\vec{w}$ to be very low as well.

c. (A) matches up with (2), since it doesn't have any EOS or BOS tags, meaning it is the first 3 words we hear as part of a single sentence, with no knowledge where these words fall within that sentence.

(B) matches up with (1), since the presence of both BOS and EOS tags indicate that it is a complete sentence.

(C) matches up with (3), since it only has a BOS tag, meaning the sentence starts with "do you think".

d. We know that under the original trigram model,

$$
\begin{aligned}
& p(w_1 w_2 w_3 w_4) \\
=& \frac{c(BOSBOSw_1)}{c(BOSBOS)} \cdot \frac{c(BOSw_1w_2)}{c(BOSw_1)} \cdot \frac{c(w_1w_2w_3)}{c(w_1w_2)} \cdot \frac{c(w_2w_3w_4)}{c(w_2w_3)} \cdot \frac{c(w_3w_4EOS)}{c(w_3w_4)} \\
=& \frac{1}{c(BOSBOS)} \cdot \frac{c(BOSw_1w_2)}{1} \cdot \frac{c(w_1w_2w_3)}{c(w_1w_2)} \cdot \frac{c(w_2w_3w_4)}{c(w_2w_3)} \cdot \frac{c(w_3w_4EOS)}{c(w_3w_4)} \cdot \\
& \quad (c(BOSBOSw_1) = c(BOSw_1) \text{ since every sentence starts with two } BOS \text{ symbols}) \\
=& \frac{1}{c(BOSBOS)} \cdot \frac{c(BOSw_1w_2) \cdot c(w_1w_2w_3) \cdot c(w_2w_3w_4) \cdot c(w_3w_4EOS)}{c(w_1w_2) \cdot c(w_2w_3) \cdot c(w_3w_4)}
\end{aligned}
$$

For the reverse trigram model, we know that

$$
\begin{aligned}
& p(w_1 w_2 w_3 w_4) \\
=& p(w_4 \mid EOS, EOS) \cdot p(w_3 \mid w_4, EOS) \cdot p(w_2 \mid w_3, w_4) \cdot p(w_1 \mid w_2, w_3) \cdot p(BOS \mid w_1, w_2) \\
=& \frac{c(w_4EOSEOS)}{c(EOSEOS)} \cdot \frac{c(w_3w_4EOS)}{c(w_4EOS)} \cdot \frac{c(w_2w_3w_4)}{c(w_3w_4)} \cdot \frac{c(w_1w_2w_3)}{c(w_2w_3)} \cdot \frac{c(BOSw_1w_2)}{c(w_1w_2)} \\
=& \frac{1}{c(EOSEOS)} \cdot \frac{c(w_3w_4EOS)}{1} \cdot \frac{c(w_2w_3w_4)}{c(w_3w_4)} \cdot \frac{c(w_1w_2w_3)}{c(w_2w_3)} \cdot \frac{c(BOSw_1w_2)}{c(w_1w_2)} \\
=& \frac{1}{c(EOSEOS)} \cdot \frac{c(BOSw_1w_2) \cdot c(w_1w_2w_3) \cdot c(w_2w_3w_4) \cdot c(w_3w_4EOS)}{c(w_1w_2) \cdot c(w_2w_3) \cdot c(w_3w_4)}
\end{aligned}
$$

But $\frac{1}{c(BOSBOS)} = \frac{1}{c(EOSEOS)}$, since all sentences in the corpus for the normal trigram model begin with two BOS symbols, and all sentences in the corpus for the reverse trigram model begin with two EOS symbols.

A more robust proof would be a proof by induction on the length n of the sentence, which goes as follows.

Let $P(n)$ be the statement that the normal trigram model and the reverse trigram model assign the same probability to every sequence $w_1 \cdots w_n$.

For the base case, let $n = 2$. Under the normal trigram model,

$$\begin{aligned} p(w_1 w_2) &= \frac{c(BOSBOSw_1)}{c(BOSBOS)} \cdot \frac{c(BOSw_1w_2)}{c(BOSw_1)} \cdot \frac{c(w_1w_2EOS)}{c(w_1w_2)} \\ &= \frac{1}{c(BOSBOS)} \cdot \frac{c(BOSw_1w_2) \cdot c(w_1w_2EOS)}{c(w_1w_2)}. \end{aligned}$$

Similarly, under the reverse trigram model,

$$\begin{aligned} p(w_1 w_2) &= \frac{c(w_2EOSEOS)}{c(EOSEOS)} \cdot \frac{c(w_1w_2EOS)}{c(w_2EOS)} \cdot \frac{c(BOSw_1w_2)}{c(w_1w_2)} \\ &= \frac{1}{c(EOSEOS)} \cdot \frac{c(BOSw_1w_2) \cdot c(w_1w_2EOS)}{c(w_1w_2)}. \end{aligned}$$

Since $c(BOSBOS) = c(EOSEOS)$, the two expressions are equal. Thus, $P(2)$ is true.

Now assume that $P(n)$ is true for some $n \geq 2$. That is, assume that both models assign $w_1 \cdots w_n$ the probability

$$\frac{1}{N} \cdot \frac{c(BOSw_1w_2) \left(\prod_{i=1}^{n-2} c(w_i w_{i+1} w_{i+2})\right) c(w_{n-1}w_nEOS)}{\prod_{i=1}^{n-1} c(w_i w_{i+1})},$$

where

$$N = c(BOSBOS) = c(EOSEOS).$$

Now consider a sequence of length $n+1$, $w_1 \cdots w_n w_{n+1}$. Adding the new word introduces the new trigram

$$w_{n-1}w_nw_{n+1},$$

and changes the final trigram from

$$w_{n-1}w_nEOS$$

to

$$w_nw_{n+1}EOS.$$

After expanding the MLE probabilities and cancelling the boundary terms where $EOSEOS = EOS$ and $BOSBOS = BOS$ just as we'd done before, both the normal and reverse trigram models reduce to

$$\frac{1}{N} \cdot \frac{c(BOSw_1w_2) \left(\prod_{i=1}^{n-1} c(w_i w_{i+1} w_{i+2})\right) c(w_nw_{n+1}EOS)}{\prod_{i=1}^{n} c(w_i w_{i+1})}.$$

Thus, the two models also assign the same probability to every sequence of length $n+1$. Therefore $P(n) \implies P(n+1)$.

Thus, by the principle of mathematical induction, the normal trigram model and the reverse trigram model assign the same probability to every finite word sequence.

5.

$$p(w_1w_2w_3w_4) = \sum_a p(a)\, p(w_1w_2w_3w_4 \mid a)$$
$$= \sum_a p(a)\, p(w_1 \mid BOS, a)\, p(w_2 \mid w_1, a)\, p(w_3 \mid w_2, a)\, p(w_4 \mid w_3, a)\, p(EOS \mid w_4, a)$$

Here, a is a latent variable that ranges over the k possible topics, which in this case are $POLITICS$, $CELEBRITIES$, $ANIMALS$ and $SPORTS$. We calculate $p(a)$ by dividing the total number of sentences of that topic in the corpus by the total size of the corpus. This guarantees that $p(w_1w_2w_3w_4) \leq 1$, since the sum of the probabilities of all k topics is equal to 1.

Also, since we assume that each sentence sticks to a single topic, we use the same latent variable a for all words in a single summation term. We sum over all topics since we don't observe the topic itself, which means we get probabilities that are way too low if we don't sum over all possible values of a.

8. **Embeddings**

a. Running the command on 50-dimensional embeddings, I got:

```
seattle: seahawks dallas atlanta wichita tacoma lauderdale florida
         spokane chino dulles
dog: dogs badger cat hound puppy dachshund sighthound poodle rat
     keeshond
communist: socialist bolshevik communists comintern trotskyist
           leftist bolsheviks cominform stalinist leninist
jpg: png svg szczepanek buteo pix gif image galleria regnum fiav
the: in its of which entire within from a part second
google: webpage yahoo web search com blogging faq editable archived
        redirection
```

Most of the examples that work "well" seem to have clear semantic categories, in that they derive most of their meaning from the word itself and not the context in which the word is used. For example, `dog` returns breeds and related animals, while `communist` returns related political terms. I also saw that proper nouns tend to return geographically or culturally related words, like `seattle` returning cities and `seahawks`. Results for `jpg` are mixed, since we see related formats like `png`, `svg`, and `gif` along with some gibberish. What we get for `the` is much less meaningful because determiners and articles are characterized more by their grammatical contexts than by a concrete meaning.

Larger values of d generally produce more precise distinctions. For example, with $d = 10$, the neighbors of `dog` include unrelated words such as `turnip`, `coronets`, and `melon`. With $d = 200$, they are mostly dog breeds, including `hound`, `mastiff`,

`spaniel`, and `terrier`. So from these observations, we can say that small vectors lose useful information, while larger vectors represent more semantic features.

b. Using the 200-dimensional embeddings, the first few results were:

```
king - man + woman:       queen, marries, isabella
paris - france + uk:      london, leeds, molview
hitler - germany + italy: mussolini, himmler, rome
child - goose + geese:    children, infants, parents
goes - eats + ate:        went, got, going
car - road + air:         helicopter, aircraft, pressurized
```

A lot of these analogies work really well: the program finds `queen`, `london`, `mussolini`, `children`, and `went`. The final analogy is not as on the nose, since we were looking for airplane, but it still moves from cars toward air vehicles. One analogy I tried was

$$\texttt{prince} - \texttt{boy} + \texttt{girl},$$

which correctly returned `princess`. Similarly, $\texttt{berlin} - \texttt{germany} + \texttt{france}$ returned `paris`. However, $\texttt{walking} - \texttt{walk} + \texttt{swim}$ did not return `swimming`, so these relationships are not always represented consistently.

The 10-dimensional versions were considerably worse. For example, $\texttt{king} - \texttt{man} + \texttt{woman}$ returned `anjou`, and $\texttt{prince} - \texttt{boy} + \texttt{girl}$ returned `spaniard`. The larger vectors have enough dimensions to encode more relationships.

I would assume that the intermediate vector $\vec{v}_{\text{king}} - \vec{v}_{\text{man}}$ represents what distinguishes a king from a generic man, particularly the royal role. Adding $\vec{v}_{\text{woman}}$ applies approximately the same relationship to a woman, producing a vector near `queen`. This works because some semantic relationships are represented by similar directions, or vector offsets, throughout the embedding space.